

MASGym: A Co-Evolving Red/Blue Security Gym for Multi-Agent LLM Systems

Secure Multi-Agent Systems — Research Idea Repository

Research concept note for collaborator discussion (not a paper)

Target Venues

- [NeurIPS \(Datasets & Benchmarks\)](#) (primary) — purpose-built environment plus task suite and metrics for a new problem regime (multi-agent security).
- [ICLR](#) — benchmark with a co-evolving red/blue methodology and an empirical finding.
- [USENIX Security](#) — the security finding (defense degradation under collusion/orchestrator compromise).
- [IEEE S&P \(Oakland\)](#) — systematized threat model and measurement across topologies.
- [AAMAS](#) — coordination, group-level emergence, and forced-coordination tasks.

Motivation

Security research on LLM agents is bottlenecked by its instruments: the strongest existing benchmarks are single-agent. ASB (arXiv:2410.02644, ICLR 2025) formalizes agent attacks/defenses but is single-agent with tools and RAG whose *only* cross-agent vector is shared memory, and it has no collusion and no compromised-orchestrator settings. AgentDojo (arXiv:2406.13352) evaluates one agent under prompt injection with deterministic, state-based `security()` checks. Yet deployed systems are increasingly orchestrated collectives: planners direct workers, agents share memory, tools are third-party, and coordination is forced. The phenomena that matter there — collusion among agents, a compromised orchestrator, malicious tools, Sybil identities, and *group-level* emergent risk — cannot be produced, let alone measured, by a single-agent harness. MASEC (arXiv:2505.02077, §4.7) states plainly that no multi-agent security benchmark exists and enumerates what one must provide. Without such an instrument, the field cannot tell whether single-agent defenses survive multi-agent deployment.

Research Gap

Established facts (corpus): (i) ToolEmu (arXiv:2309.15817, ICLR 2024) is an LM-emulated sandbox in which an LLM emulates arbitrary tool execution from specifications, paired with an adversarial emulator and LM safety/helpfulness evaluators at human agreement $\kappa \approx 0.48$. (ii) ASB (arXiv:2410.02644, ICLR 2025) introduces $\text{NRP} = \text{PNA} \cdot (1 - \text{ASR})$ but is single-agent (tools + RAG), with shared memory as its lone cross-agent vector, and *no* collusion and *no* compromised-orchestrator settings. (iii) AgentDojo (arXiv:2406.13352) uses deterministic state-based `security()` checks rather than an LLM judge, so a successful injection cannot hijack the evaluator. (iv) MASEC (arXiv:2505.02077, §4.7) states no multi-agent security benchmark exists

and asks for heterogeneous roles, partial observability, forced coordination, co-evolving red/blue, and group-level emergence metrics. Unresolved: an environment that *natively* parameterizes collusion, orchestrator/tool compromise, and Sybils, and evaluates risk at the *system* level. Whether single-agent defenses degrade there is the open question we target and mark (*hypothesis*).

System and Threat Model

Because MASGym is an environment, its threat model is *parameterized* and configurable, not fixed.

- **Roles / agents:** heterogeneous roles (orchestrator, workers, tool-executors, shared-memory store); 3–50 agents. Partial observability: agents see only their local context and permitted messages.
- **Topology / coordination:** star, chain, tree, and mesh; forced-coordination tasks that no single agent can complete alone.
- **Trust / oversight:** per-agent trust boundaries; optional human-approval hooks; orchestrator may be trusted or compromised (a switch).
- **Attacker (parameterized):** adversary fraction over agents; independent vs. colluding; static vs. adaptive; compromised orchestrator on/off; malicious tools; Sybil identities; poisoned shared memory. Objective: system-level compromise (unsafe action, exfiltration, coordination sabotage).
- **Trusted computing base:** the deterministic state checker and the environment harness are trusted; emulated tools have no real side effects. **Out of scope:** real-world tool side effects, physical actuation, model-weight attacks.

Core Idea

MASGym is, to our knowledge, the first benchmark/environment purpose-built for *multi-agent* security. It provides forced-coordination tasks, heterogeneous roles, and partial observability, plus *native* parameterized settings for colluding agents, compromised orchestrators, malicious tools, and Sybils. It evaluates risk at the *system* level rather than per step, runs a *co-evolving red/blue* protocol, and reports a *system-level NRP* that generalizes ASB’s $NRP = PNA \cdot (1 - ASR)$. It is built on ToolEmu-style LM emulation for breadth of tools and AgentDojo-style deterministic state-based checks for evaluation integrity (so a successful injection cannot hijack the judge). To avoid the “benchmark without a finding” critique, MASGym ships with a concrete scientific finding as its headline result: that defenses effective in single-agent ASB/AgentDojo degrade under collusion and compromised-orchestrator conditions (*hypothesis*). This is an environment/methodology contribution, not an attack or defense per se.

Novelty

MASGym is *fundamentally multi-agent* because it exists to *isolate* phenomena that single-agent harnesses cannot reproduce. It is not ASB-applied-to-many: ASB’s only cross-agent vector is shared memory, and it has no collusion and no compromised-orchestrator settings, so re-running ASB N times does not create coordinated adversaries, orchestrator trust, or group-level emergence. It is not AgentDojo-at-scale: AgentDojo is single-agent, and its (excellent) deterministic checker was designed for one agent’s state. It is not a *model-swap* or *prompt-filter* study, and not mere *engineering*: the scientific object is the controlled variation of *coordination*, *collusion*, and *orchestrator* variables and the system-level metrics that quantify their effect — variables that are undefined in

a single-agent setting. Forced coordination, collusion, compromise propagation, and Sybil pressure are group-level by construction.

Expected Contributions

- An open environment and task suite for multi-agent LLM security with heterogeneous roles, partial observability, and forced coordination.
- A parameterized adversary model: adversary fraction, collusion on/off, adaptivity, orchestrator compromise, malicious tools, Sybils, poisoned memory.
- System-level metrics, including a system-level NRP generalizing ASB’s, that measure risk over the collective rather than per step.
- A co-evolving red/blue protocol as a reusable evaluation methodology.
- A headline empirical *finding* on single-agent-defense degradation under collusion/orchestrator compromise (*hypothesis*), so the benchmark ships with a result.

Preliminary Technical Direction

Architecture: (1) an emulation layer where an LLM emulates tool execution from specs (ToolEmu-style), with an adversarial-emulator mode; (2) a deterministic state checker that evaluates system-level security predicates over the global state trace (AgentDojo-style), never an LLM judge on the critical path, so injections cannot hijack scoring; (3) an orchestration layer that instantiates topology, roles, observability, and the adversary parameters. Define a system-level generalization $NRP_{sys} = PNA_{sys} \cdot (1 - ASR_{sys})$, where ASR_{sys} is measured against system-level security predicates and PNA_{sys} is benign coordinated-task performance; the exact aggregation is a design contribution to be validated, not assumed. The co-evolving loop alternates a red policy (proposes attack configurations from the parameter space) and a blue policy (proposes defenses), logging a trajectory of (attack, defense) pairs and their system-NRP. Emulation fidelity is bounded by ToolEmu’s reported $\kappa \approx 0.48$; we surface this as a first-class caveat rather than treating emulator verdicts as ground truth.

Evaluation Strategy

This is a benchmark, so “evaluation” means validating the instrument and demonstrating the headline finding. **Models:** open (Llama-3, Qwen2.5, Mistral) and proprietary (GPT-4o, Claude-3.5) agents. **Scale/topology:** 3–50 agents across star, chain, tree, and mesh. **Adversary sweeps:** adversary fraction; independent vs. colluding; static vs. adaptive; compromised orchestrator on/off; malicious tools; Sybils; poisoned memory. **Metrics:** system-level ASR, collusion success rate, compromise propagation rate, coordination quality, system-NRP, time-to-detection, recovery rate, token/API and latency cost, and scalability with agent count. **Baselines:** ASB re-run in single-agent mode; AgentDojo; no-defense; per-agent defenses lifted into the multi-agent setting. **Ablations:** collusion on/off; orchestrator trusted vs. compromised; emulated vs. real tools; deterministic vs. LLM-judge evaluation (to quantify evaluator-hijack risk). The environment is instrument, not verdict: we never equate a lower system-ASR or higher system-NRP with “secure”; results characterize measured risk under stated configurations and are (*hypothesis*) until produced.

Potential Risks

Emulation fidelity is the central threat to validity: ToolEmu’s human agreement is $\kappa \approx 0.48$, so emulator-derived outcomes carry noise; we mitigate by routing scoring through the deterministic checker and reporting the emulated-vs-real-tools ablation. The “benchmark-without-a-finding” critique is mitigated by shipping the degradation finding, but that finding must actually replicate across models and topologies (*hypothesis*); if defenses do *not* degrade, the contribution reframes as a (still valuable) negative result. API/compute cost grows with agent count and the co-evolving loop, constraining the reachable region of the parameter space. Construct-validity risk: the system-NRP aggregation must be justified, not asserted, or reviewers will discount it.

Publication Readiness

Confidence: **8/10**. The gap is explicit — MASEC (§4.7) states no multi-agent security benchmark exists — and the design reuses established, credible components (ToolEmu emulation, AgentDojo deterministic checks, ASB’s NRP) in a genuinely new regime, which fits NeurIPS D&B and ICLR strongly, with USENIX/S&P for the finding and AAMAS for coordination. The score is held below 9 by emulation-fidelity limits ($\kappa \approx 0.48$), compute cost, and dependence of the headline finding on replication; all quantitative outcomes remain (*hypothesis*) until measured.